

Rajiv Gandhi Proudyogiki Vishwavidyalaya Bhopal

M.Tech (Mining Engineering)

Third Semester Syllabus

MTMI 301(A) - SAFETY AND RISK MANAGEMENT IN MINES

Introduction to Safety

Concept of Safety, Importance of industrial safety, Safety scenario in Indian mines

Accident Analysis and Control

Concept of an accident, accident classification, Causes of accidents; Domino sequence, Unsafe acts and conditions; Safety performance measures. Documentation of accidents, Accident investigation and analysis; Role of supervisory staff and safety committee for accident prevention.

Hazard, Risk Issues and Hazard Assessment

Concept of hazard, hazard monitoring, assessment; Risk management, Stochastic models for accident prediction, Consequence modeling. Preliminary hazard analysis (PHA), Hazard operability studies (HAZOP), Safety warning systems. Safety Management Plan

Behavioural Safety and Safety Audit

Human behavioural study; Human error analysis; Components of safety audit, Audit methodology, Non conformity reporting (NCR), Audit Checklist and report.

Safety Management and Systems Safety

Principles of safety management – Workplace design, Development of safety culture, Safety Barrier; Promoting Safety at workplace. Systems engineering approach to safety management.

MTMI 301(B) - RELIABILITY ENGINEERING

Basic Concepts of Reliability

Introduction to Reliability Engineering. Failures and failure modes. Causes and mechanism of failures. Maintainability and availability.

Component Reliability and Hazard Models

Failure distributions. Constant and Time dependent hazard models. Exponential, Normal, Lognormal and Weibull failure models.

Modelling of System Reliability

Reliability block diagrams. Series and parallel configurations, Redundancies. Markov Chain.

Design for Reliability

Reliability allocation. Designing for higher reliability. Failure analysis: FMEA, FMECA, FTA. Reliability optimization.

Equipment Reliability Analysis

Type of failure data. Sources of failure data. Classification of failure data. Identification of Failure distributions. Goodness-of-fit Test. Analysis of equipment reliability.

MTMI 301(C) -Underground Mine Planning

Mining Industry in India: An Overview Characteristics of Planning Process Scope of mining activities.

Stages of mine planning. Feasibility and Project Reports Contents, preparation and evaluation.

Preparation of mine plan and mine environmental plan. Capacity of a Mine Delineation of mining area. Annual output and life of the mine.

Mine Entries Opening of single and multiple seams/veins at various inclinations – Type (shaft, incline or adit), number, location and design.

Division of Mining Area Division of the mining area into working units on district and level pattern. Dimensions of panels and blocks. Production Planning and Scheduling

MTMI 302(A) - SUBSIDENCE ENGINEERING

Theories of surface and sub-surface subsidence due to mining and non-mining causes. Zones of movement in the overlying beds. Rock kinematics.

Types of subsidence. Factors affecting subsidence. Methods of prediction of subsidence- empirical, analogue, numerical and physical models.

Prediction of subsurface subsidence and subsidence monograms. Measurement of subsidence. Time dependent component of subsidence.

Special mining layouts to minimize subsidence's. Impact of subsidence on structures. Design of shaft and safety pillars.

MTMI 302(B) - ROCK SLOPE ENGINEERING

Role of Slope Stability in Economic Design and Operation of Open Pit Mines
Site characterization
Structural geology and site investigation, strength properties of rock and soil

Types and Mechanics of Slope Failure. Types and mechanics of slope failure: plane, wedge, toppling and circular and non-circular,
Factors Affecting Slope Stability, Geological factors, slope geometry, ground water, equipment loading, dynamic loading and effect of time.

Slope Stability Analysis: Methods of slope stability analysis. Limit equilibrium and Numerical analyses of rock and soil slopes. Deterministic and probabilistic approaches. Dynamic simulation and Seepage Analysis.

Instrumentation and monitoring: Types of slope movement, methods of Field instrumentation and monitoring. Conventional and advance monitoring.

Stabilization of slope: Stabilization and strengthening of rock slopes and dump slope, Selection of stabilization measures, various types of methods for strengthening of slopes, Design of Waste Dumps and Tailings Dams, Design of retaining wall.

MTMI 302(C) - WASTE MANAGEMENT IN MINES

Environmental and global issues related with mining metal, minerals atomic minerals and fossil fuels.

Emergency and best available techniques for management of waste rock, tailings and water effluents.

Chemical aspects of environmental pollution by mine wastes and their impact. Production and characterization of solid wastes in different types of mines.

Generation and characterization of mine effluents and leachate. Management of different types of mine wastes: Integrated solid waste management.

Tailings – characterization, technical issues, sampling and analysis, site selection and design of tailings impoundment, tailings dam failure.